

Probability Theory: Class Test Problems

Undergraduate Practice Set

Sigma-Field Validation

Determine whether each given collection $\mathcal{F}$ is a valid σ -algebra on the indicated sample space Ω . Justify your answer.

1. $\Omega = \{1, 2, 3\}$, $\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$.
2. $\Omega = \{a, b, c, d\}$, $\mathcal{F} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c, d\}, \Omega\}$.
3. $\Omega = \mathbb{N}$ (natural numbers), $\mathcal{F} =$ all finite subsets of $\mathbb{N}$ together with their complements.
4. $\Omega = \mathbb{R}$, $\mathcal{F} = \{\emptyset, \mathbb{R}, (-\infty, 0], (0, \infty)\}$.
5. $\Omega = \{1, 2, 3, 4\}$, $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 3\}, \{2, 4\}, \Omega\}$.
6. $\Omega = [0, 1]$, $\mathcal{F} = \{A \subseteq [0, 1] : A \text{ is countable or } A^c \text{ is countable}\}$.

Classification of Sample Spaces

For each sample space below, state whether it is: (A) Discrete finite (B) Continuous finite (bounded interval in $\mathbb{R}$) (C) Countably infinite (D) Uncountably infinite (including continuous infinite, e.g., $\mathbb{R}$)

Also, if the phrase “Uncountable finite” appears — note: this is a trick; uncountable implies infinite, so “uncountable finite” is impossible. Correct any such misnomer.

7. Rolling a fair six-sided die once.
8. Measuring the exact height (in cm) of a randomly selected student, where height $\in [140, 220]$.
9. The number of coin tosses until the first head appears.
10. The set of all possible outcomes of flipping a coin infinitely many times.
11. The possible temperatures (in Kelvin) of an ideal gas in a fixed container, ranging from 0 K to 1000 K.
12. The number of goals scored in a soccer match (no upper bound).
13. Choosing a random point inside the unit square $[0, 1] \times [0, 1]$.
14. The number of emails received in a day (maximum 10^6 due to server limit).

Valid Probability Space (Axiomatic Definition)

For each triple $(\Omega, \mathcal{F}, P)$, verify whether it satisfies Kolmogorov’s axioms. If not, state which axiom(s) fail.

Axioms: 1. $P(A) \geq 0$ for all $A \in \mathcal{F}$. 2. $P(\Omega) = 1$. 3. For any countable collection of disjoint sets $A_1, A_2, \dots \in \mathcal{F}$, $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

15. $\Omega = \{H, T\}$, $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\}$, $P(\{H\}) = 0.6$, $P(\{T\}) = 0.4$.
16. $\Omega = \{1, 2, 3\}$, $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3\}, \Omega\}$, $P(\{1, 2\}) = 0.5$, $P(\{3\}) = 0.5$.
17. $\Omega = \mathbb{N}$, $\mathcal{F} = 2^{\Omega}$ (all subsets), $P(\{k\}) = \frac{6}{\pi^2 k^2}$, and for any A , $P(A) = \sum_{k \in A} P(\{k\})$.

18. $\Omega = [0, 1]$, $\mathcal{F} = \text{Borel } \sigma\text{-algebra}$, $P(A) = \lambda(A)$ (Lebesgue measure). Check if this is valid.
19. $\Omega = \{0, 1\}$, $\mathcal{F} = \{\emptyset, \Omega\}$ (trivial σ -algebra), $P(\emptyset) = 0$, $P(\Omega) = 2$ (is this valid?).
20. $\Omega = \{1, 2, 3\}$, $\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$, $P(\{1\}) = 0.5$, $P(\{2, 3\}) = 0.5$, $P(\Omega) = 1$, $P(\emptyset) = 0$.
21. $\Omega = \mathbb{R}$, $\mathcal{F} = \{\emptyset, \mathbb{R}\}$, $P(\emptyset) = 0$, $P(\mathbb{R}) = 1$.
22. $\Omega = \{a, b, c\}$, $\mathcal{F} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c\}, \{a, c\}, \{b, c\}, \Omega\}$, $P(\{a\}) = 2$, $P(\{b\}) = -1$, $P(\{c\}) = 0$, extending additively.
23. $\Omega = \{1, 2, 3, \dots\}$, $\mathcal{F} = 2^\Omega$, $P(\{i\}) = \frac{1}{2^i}$ for $i \geq 1$, and $P(A) = \sum_{i \in A} \frac{1}{2^i}$.

Definition Reminder

A function $X : \Omega \rightarrow \mathbb{R}$ is a **random variable** (measurable) with respect to a σ -algebra $\mathcal{F}$ on Ω if for every Borel set $B \subseteq \mathbb{R}$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Equivalently, it is enough to check that for every $x \in \mathbb{R}$, $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$.

Problem 0.1: 1

Let $\Omega = \{1, 2, 3\}$ with $\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$. Define $X(1) = 5$, $X(2) = 5$, $X(3) = 10$. Is X a valid random variable?

Answer

We check the inverse image of Borel sets. X takes only values 5 and 10. - $X^{-1}(\{5\}) = \{1, 2\}$. Is $\{1, 2\} \in \mathcal{F}$? No (only $\{1\}$ and $\{2, 3\}$ are present). - Hence X is not measurable. **Conclusion:** Not a valid random variable.

Problem 0.2: 2

Same Ω and $\mathcal{F}$ as in Problem 1. Define $Y(1) = 0$, $Y(2) = 1$, $Y(3) = 1$. Is Y a random variable?

Answer

Y takes values 0 and 1. - $Y^{-1}(\{0\}) = \{1\} \in \mathcal{F}$ - $Y^{-1}(\{1\}) = \{2, 3\} \in \mathcal{F}$ - For any Borel set B , $Y^{-1}(B)$ is either $\emptyset$, $\{1\}$, $\{2, 3\}$, or Ω , all in $\mathcal{F}$. **Conclusion:** Yes, Y is a valid random variable.

Problem 0.3: 3

$\Omega = \mathbb{N}$ (natural numbers), $\mathcal{F} = 2^\mathbb{N}$ (all subsets). Is *every* function $X : \mathbb{N} \rightarrow \mathbb{R}$ a random variable?

Answer

Yes. Because $\mathcal{F} = 2^\mathbb{N}$ contains every subset of Ω . For any X , $X^{-1}(B)$ is some subset of $\mathbb{N}$, hence belongs to $\mathcal{F}$. Thus every function is measurable. **Conclusion:** Yes.

Problem 0.4: 4

$\Omega = \mathbb{R}$, $\mathcal{F} = \text{Borel } \sigma\text{-algebra on } \mathbb{R}$. Define $X(\omega) = \omega^2$. Is X a random variable?

Answer

For any $x \in \mathbb{R}$,

$$\{\omega : X(\omega) \leq x\} = \begin{cases} \emptyset & \text{if } x < 0, \\ [-\sqrt{x}, \sqrt{x}] & \text{if } x \geq 0. \end{cases}$$

In both cases this set is a Borel set (closed interval or empty). Therefore X is measurable.

Conclusion: Yes, X is a random variable.

Problem 0.5: 5

$\Omega = \mathbb{R}$, $\mathcal{F} = \{\emptyset, \mathbb{R}\}$ (trivial σ -algebra). Let $X(\omega) = 0$ for all ω . Is X a random variable?

Answer

For any Borel set B : - If $0 \in B$, then $X^{-1}(B) = \mathbb{R} \in \mathcal{F}$. - If $0 \notin B$, then $X^{-1}(B) = \emptyset \in \mathcal{F}$. Hence X is measurable. **Conclusion:** Yes (constant functions are always measurable).

Problem 0.6: 6

Same Ω and $\mathcal{F}$ as in Problem 5. Define $Y(\omega) = \omega$. Is Y a random variable?

Answer

Take $B = \{0\}$ (a Borel set). Then $Y^{-1}(\{0\}) = \{0\}$. Is $\{0\}$ equal to $\emptyset$ or $\mathbb{R}$? No. Thus $\{0\} \notin \mathcal{F}$. So Y is not measurable. **Conclusion:** No.

Problem 0.7: 7

$\Omega = [0, 1]$, $\mathcal{F}$ = Borel σ -algebra on $[0, 1]$. Define $X(\omega) = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}(\omega)$ (indicator of rationals). Is X a random variable?

Answer

X takes values 0 and 1. - $X^{-1}(\{1\}) = \mathbb{Q} \cap [0, 1]$. - $X^{-1}(\{0\}) = [0, 1] \setminus \mathbb{Q}$ (irrationals). Both sets are Borel sets (rationals are F_σ , irrationals are G_δ). Hence measurable. **Conclusion:** Yes.

Problem 0.8: 8

$\Omega = \{1, 2, 3, 4\}$, $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$. Define $X(1) = X(2) = 0$, $X(3) = X(4) = 1$. Is X a random variable?

Answer

$X^{-1}(\{0\}) = \{1, 2\} \in \mathcal{F}$, $X^{-1}(\{1\}) = \{3, 4\} \in \mathcal{F}$. Any other Borel preimage is $\emptyset$ or Ω , both in $\mathcal{F}$. **Conclusion:** Yes.

Problem 0.9: 9

Same Ω and $\mathcal{F}$ as Problem 8. Define $Y(1) = 0$, $Y(2) = 1$, $Y(3) = 0$, $Y(4) = 1$. Is Y a random variable?

Answer

$Y^{-1}(\{0\}) = \{1, 3\}$. Is $\{1, 3\} \in \mathcal{F}$? No (the only non-trivial sets are $\{1, 2\}$ and $\{3, 4\}$). Thus Y is not measurable. **Conclusion:** No.

Problem 0.10: 10

$\Omega = (0, 1)$, $\mathcal{F}$ = Borel σ -algebra on $(0, 1)$. Define $X(\omega) = \frac{1}{\omega}$. Is X a random variable? (Note: X maps to $(1, \infty)$.)

Answer

For any $x \in \mathbb{R}$,

$$\{\omega : X(\omega) \leq x\} = \begin{cases} \emptyset & \text{if } x \leq 1, \\ (0, \frac{1}{x}] & \text{if } x > 1. \end{cases}$$

Each set is a Borel set (empty or half-open interval). Hence measurable. **Conclusion:** Yes.

Problem 0.11: 11

$\Omega = \mathbb{R}$, $\mathcal{F}$ = countable/co-countable σ -algebra: $A \in \mathcal{F}$ iff A is countable or A^c is countable. Define $X(\omega) = 0$ for all ω . Is X measurable?

Answer

Constant function: $X^{-1}(B) = \mathbb{R}$ if $0 \in B$, else $\emptyset$. Both $\mathbb{R}$ and $\emptyset$ are in $\mathcal{F}$ ($\mathbb{R}$ is co-countable, $\emptyset$ is countable). **Conclusion:** Yes.

Problem 0.12: 12

Same Ω and $\mathcal{F}$ as Problem 11. Define $X(\omega) = \omega$. Is X measurable?

Answer

Take $B = \{0\}$. $X^{-1}(\{0\}) = \{0\}$, which is countable, hence in $\mathcal{F}$. Good. But take $B = [0, 1]$ (uncountable interval). $X^{-1}([0, 1]) = [0, 1]$. This set is uncountable and its complement $(-\infty, 0) \cup (1, \infty)$ is also uncountable. Hence $[0, 1] \notin \mathcal{F}$. Thus X is not measurable. **Conclusion:** No.

Problem 0.13: 13

Let $\Omega = \{H, T\}^{\mathbb{N}}$ (infinite coin tosses) with $\mathcal{F}$ the product σ -algebra (generated by cylinder sets). Define $X(\omega)$ = the number of heads in the first 3 tosses. Is X measurable?

Answer

X takes values $\{0, 1, 2, 3\}$. For each k , $\{\omega : X(\omega) = k\}$ is a finite union of cylinder sets (e.g., $k = 0$ corresponds to TTT followed by anything). Cylinder sets are in $\mathcal{F}$ and finite unions remain in $\mathcal{F}$. Hence each preimage is in $\mathcal{F}$. **Conclusion:** Yes.

Problem 0.14: 14

$\Omega = \mathbb{R}$, $\mathcal{F}$ = Borel σ -algebra. Define $X(\omega) = \begin{cases} 1 & \text{if } \omega \in \mathbb{Q} \\ 0 & \text{if } \omega \notin \mathbb{Q} \end{cases}$. Show X is a random variable.

Answer

Same as Problem 7 essentially. $X^{-1}(\{1\}) = \mathbb{Q}$ (Borel), $X^{-1}(\{0\}) = \mathbb{R} \setminus \mathbb{Q}$ (Borel). So measurable. **Conclusion:** Yes.

Problem 0.15: 15

Let $\Omega = \{1, 2, \dots, 10\}$, $\mathcal{F} = \{\emptyset, \{1, 3, 5, 7, 9\}, \{2, 4, 6, 8, 10\}, \Omega\}$ (odds and evens). Define $X(1) = X(3) = X(5) = X(7) = X(9) = 1$, $X(2) = X(4) = X(6) = X(8) = X(10) = 0$. Is X a random variable?

Answer

$X^{-1}(\{1\}) = \{1, 3, 5, 7, 9\} \in \mathcal{F}$, $X^{-1}(\{0\}) = \{2, 4, 6, 8, 10\} \in \mathcal{F}$. All other preimages are $\emptyset$ or Ω , both in $\mathcal{F}$. **Conclusion:** Yes.

Problem 0.16: 16

Same $\Omega, \mathcal{F}$ as Problem 15. Define $Y(1) = 1$, $Y(2) = 2$, $Y(3) = 1$, $Y(4) = 2$, $Y(5) = 1$, $Y(6) = 2$, $Y(7) = 1$, $Y(8) = 2$, $Y(9) = 1$, $Y(10) = 2$. Is Y measurable?

Answer

$Y^{-1}(\{1\}) = \{1, 3, 5, 7, 9\}$ (odds) $\in \mathcal{F}$, $Y^{-1}(\{2\}) = \{2, 4, 6, 8, 10\}$ (evens) $\in \mathcal{F}$. Hence yes. **Conclusion:** Yes.

Problem 0.17: 17

Let $\Omega = [0, 1]$, $\mathcal{F}$ = Borel σ -algebra. Define $X(\omega) = \sin(2\pi\omega)$. Is X a random variable?

Answer

X is continuous. For any x , $\{\omega : X(\omega) \leq x\}$ is a Borel set (it is the preimage of $(-\infty, x]$ under a continuous function, hence Borel). Therefore X is measurable. **Conclusion:** Yes.

Problem 0.18: 18

$\Omega = \mathbb{R}$, $\mathcal{F}$ = Borel σ -algebra. Define $X(\omega) = \omega \cdot \mathbf{1}_{\mathbb{Q}}(\omega)$. Is X measurable? (Hint: $X(\omega) = 0$ on irrationals, $X(q) = q$ for rationals.)

Answer

For any $x \in \mathbb{R}$, $\{\omega : X(\omega) \leq x\} = \mathbb{Q} \cap (-\infty, x] \cup \{\text{irrationals}\}$? Actually: - If ω is rational and $\omega \leq x$, then $X(\omega) = \omega \leq x$. - If ω is irrational, $X(\omega) = 0 \leq x$ whenever $0 \leq x$? Wait: For $x < 0$, irrationals give $0 > x$ so they are NOT included. So careful:

Case $x < 0$: Only rationals $\leq x$ (since irrationals give 0 which is $> x$). So set $= (-\infty, x] \cap \mathbb{Q}$ — countable, hence Borel.

Case $x \geq 0$: All irrationals satisfy $X = 0 \leq x$, so they are included. Also rationals $\leq x$. So set $= (-\infty, x] \cup \{\text{irrationals}\}$? No, union is whole real line. Actually the set is $(-\infty, x] \cup \{\text{irrationals}\}$? But irrationals are already partly in $(-\infty, x]$. Let's write: Set $= \{\omega \in \mathbb{R} : X(\omega) \leq x\} = \{\omega \in \mathbb{Q} : \omega \leq x\} \cup \{\omega \notin \mathbb{Q} : 0 \leq x\}$.

If $x \geq 0$, the irrational condition $0 \leq x$ is always true, so all irrationals included. Thus set $= \mathbb{R} \setminus \{\text{rationals} > x\}$. That's a Borel set (complement of a countable set). If $x < 0$, set $= \{\text{rationals} \leq x\}$ (countable), Borel.

Thus X is measurable. **Conclusion:** Yes.

Summary of Key Ideas

- A function is a random variable iff preimages of Borel sets lie in the σ -algebra.
- On a discrete space with power set σ -algebra, every function works.

- On a coarse σ -algebra (e.g., trivial, odds/evens), only functions constant on the atoms of $\mathcal{F}$ are measurable.
- Continuous functions on $\mathbb{R}$ with Borel σ -algebra are always measurable.
- Indicator functions of measurable sets are measurable.

Bonus Conceptual Questions

24. Is it possible to have a probability space where Ω is countable infinite, $\mathcal{F} = 2^\Omega$, and $P(\{\omega\}) = 0$ for every $\omega \in \Omega$? Justify using countable additivity.
25. Give an example of a sample space that is “uncountable finite” or argue why such a thing cannot exist.